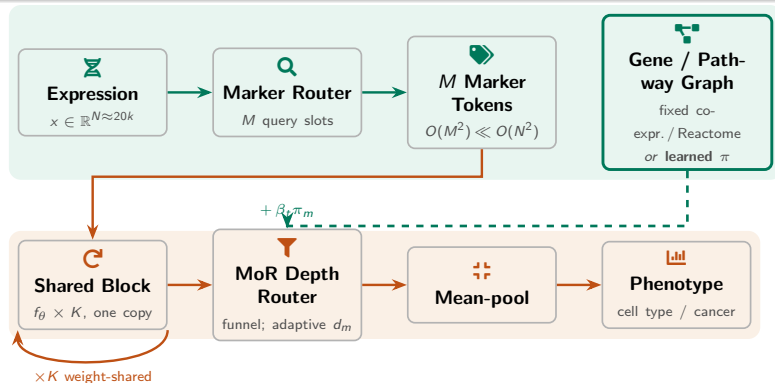


Problem & design: a biology-informed recursive transformer

Problem statement

Classify a cell/sample from $x \in \mathbb{R}^N$ ($N \approx 20k$ genes) *accurately & cheaply*. Two costs are our targets: **(T) token cost** — attention is $O(N^2)$ in genes; **(F) FLOP/depth cost** — a K -layer stack is $O(K)$ FLOPs and $O(K)$ params, spent uniformly on every token. **Goal: (T)** $O(N^2) \rightarrow O(M^2)$, $M \ll N$; **(F)** $O(K) \rightarrow$ **adaptive per-token depth** at **constant params** — guided by biology.



A marker selection: $N \rightarrow M$ tokens, $O(M^2)$ attention • **B** one shared block $\times K$ + **MoR** adaptive depth • **prior** π nudges **hubs** deeper (label-free)

Results: learned routing wins; efficiency holds; fixed priors don't

Token reduction

macro-F1 vs # marker tokens M (13-dataset suite)

	$M=16$	$M=64$	$M=128$	$M=256$
Baron	33.7	53.9	62.6	66.9
Lung	30.8	61.5	71.4	78.4
T-cell	25.5	43.1	47.4	61.5
Prostate	74.3	78.0	76.7	77.3

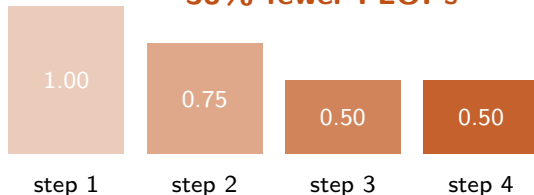
Vanilla vs Recursive vs MoR @ $K=4$

suite mean (13)	Vanilla	Recursive	MoR
Params (unique)	299,520	74,880	75,168
FLOPs (rel.)	1.00×	1.00×	0.62×
Compute saved	—	—	38%
macro-F1	51.4±16.8	52.3±17.1	50.4±17.1
accuracy	66.2	67.1	66.6

Equal accuracy: MoR = 4× fewer params than Vanilla + 38% fewer FLOPs

Adaptive depth $\Rightarrow$ compute funnel

mean depth 2.75/4
 $\sim$ 38% fewer FLOPs



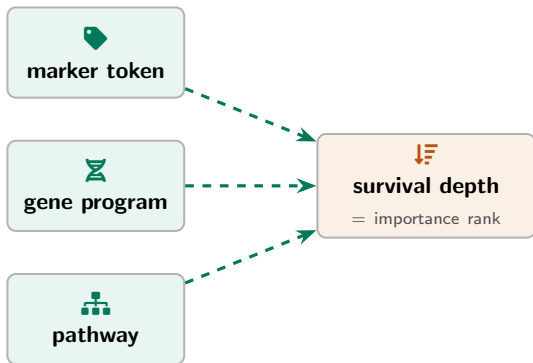
Verdicts (multi-seed, 13 datasets)

+ **Learned** routing graph: +12.0 F1 single cell (66.5 vs 54.5), the decisive win — beats fixed biology, random and no prior.

+ Adaptive depth: $\sim$ 38% compute saved at matched accuracy & params.

— Fixed biology prior: = its random-graph control, **collapses** on some suites; calibration no gain

Biologically meaningful token reduction — and where to push next



Each token $\approx$ a learned **signature / gene program** — a deployable **marker panel**, not a black box. A token's **recursion depth** is a compute-based **importance score**.

Negatives $\Rightarrow$ directions

- **Stronger injection:** prior is an additive, annealed-away bias $\rightarrow$ use **FiLM/gating** so centrality *modulates* routing per cell.
- **Better graph:** swap co-expression centrality for **directed GRNs / perturbation graphs** (causal, not correlational).
- **Right granularity:** route **pathway tokens** $\Rightarrow$ "depth per pathway" = compute a biological process needs.
- **Validate:** check survival-depth rankings against **known lineage / disease markers**.

Take-away: token reduction is the biologically meaningful lever that *works* — and a **learned** routing graph (next slide) turns the biology-prior direction into a real accuracy gain.

Why a *learned* graph beats a fixed biological one

The router smooths expression along a gene graph A :

$$\tilde{x} = (1 - \lambda)x + \lambda Ax.$$

Fixed A (co-expression / Reactome):

- label-free $\Rightarrow \partial \mathcal{L} / \partial A = 0$ (edges never learn);
- annealed $\Rightarrow$ converged model $\equiv$ none;
- a degree-matched *shuffle* gives the same generic smoothing: $\mathbb{E}[\text{bio}] = \mathbb{E}[\text{random}]$ (FACT B).

Learned $A = \tilde{E}\tilde{E}^\top$ (rank r , embedding E):

$$\tilde{x} = (1 - \lambda)x + \lambda(x\tilde{E})\tilde{E}^\top, \quad \frac{\partial \mathcal{L}}{\partial E} \neq 0.$$

Edges are chosen to cut the loss. Low rank $\Rightarrow r$ latent **gene programs**; propagation denoises each gene toward its program's

One patient, two co-varying markers

Genes g_1, g_2 : weak, *noisy*, but co-vary with the label.

Learned graph learns $\tilde{E}_{g_1} \approx \tilde{E}_{g_2}$ (same program):

$$\tilde{x}_{g_1} \approx \frac{1}{2}(x_{g_1} + x_{g_2})$$

independent noise averages down ($\times \sqrt{2}$ **SNR**),
shared signal kept $\Rightarrow$ a **cleaner marker token**.

Random / fixed graph links g_1 to *unrelated* genes $\Rightarrow$
averages *in* noise $\Rightarrow$ over-smooths, no SNR gain.

On data (10 seeds): learned – none = **+12.0** F1 on
single cell (66.5 vs 54.5; up to **+19** on Baron /
T-cell) and **+1.9** on P-NET (**+5.7** prostate). Beats
co-expression, Reactome *and* random — the first
mechanism that does.